Danyal Gu

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SKILLS

Mechanical Design & Simulation: PTC Creo, SolidWorks, FEA (ANSYS), CFD (Pointwise), DFMA
Drafting & PDM: AutoCAD, GD&T, PTC Windchill PDM, Bluebeam Revu
Technical Tools: MS Office, MATLAB, Excel VBA, C++, Java, Python, Microcontrollers, 3D Printing
Manufacturing: Fixture Design, Manual Milling, Lathe Machining, 3D Printing

EXPERIENCE

Mechanical Design Co-op May 2026 - Aug 2026
TK Modular Systems *Oakville, ON*

- Modified 3D CAD assemblies in PTC Creo for two large-scale biopharmaceutical water purification skid systems (40+ skids each), interpreting P&IDs to implement client-requested structural and piping design changes
- Created 150+ new and revised 300+ existing fabrication, spool, conduit, and structural frame drawings spanning 4,000+ components, applying GD&T to produce actionable shop-floor documentation
- Managed the drawing release lifecycle in Windchill PDM, resolving cross-functional markups from mechanical and process engineering reviews against client specifications and internal design standards

Undergraduate Research Assistant Sep 2026 - Present
University of Waterloo (Multi-Physics Interaction Lab) *Waterloo, ON*

- Selected as a URA through the President's Research Award for Prof. Jean-Pierre Hickey's rotating detonation engine (RDE) project, targeting 1,000 lbf thrust as the second Canadian university-led RDE program
- Built a database from 30+ RDE papers, logging fuel, oxidizer, equivalence ratio, injection style, wave velocity, pressure and thrust data to inform design parameters ahead of CFD/HPC work

Senior Chassis Member Sep 2025 - Present
University of Waterloo Formula SAE (FSAE) *Waterloo, ON*

- Designed a brake line mounting system around one standardized fixture used at all 20 mounting points along the carbon fiber floor, resolving critical packaging constraints to ensure safe line retention
- Developed a 3D-printed welding fixture using in-context assembly references, creating a mechanical hard-stop to position the master switch below the helmet impact zone per FSAE safety rules, achieving <1mm vertical accuracy and cutting setup time to 5 minutes with zero post-weld adjustments
- Machined interference-fit assemblies and structural inserts from aluminum and steel stock, utilizing datum alignment on manual mills and lathes to hold strict dimensional tolerances within ± 0.01mm

PROJECTS

FSAE 2026 Chassis Torsional Stiffness Analysis *ANSYS Static Structural, SolidWorks, Excel*

- Built a finite element beam spaceframe model in ANSYS Static Structural under a 600Nm torsional load, using spherical joints to replicate suspension kinematics, with accumulator and AIP added
- Evaluated four frame configurations (baseline, with AIP, with accumulator, and combined) identifying that the accumulator adds 25.6% torsional stiffness and the full configuration achieves 2660 Nm/deg, a 31% improvement over the baseline frame
- Built an Excel tool to post-process stiffness results and designed a rigid fixture to interface the chassis with the physical torsion rig for FEA correlation

Gantry-Driven Checker Robot *SolidWorks, 3D Printing, C++*

- Designed the complete mechanical assembly in SolidWorks (gravity-fed hopper, rotating gate dispenser, and 3-axis rack and pinion gantry system) and led full physical assembly of the prototype
- Selected filament materials based on mechanical requirements, PETG for structural racks, carbon fiber for high-stress pinions, PLA for tolerance-critical sensor enclosures
- Tested over 10 trials, reaching 94% colour detection accuracy and 90% mechanical reliability; traced the 75% placement accuracy to gantry frame deflection and proposed guide rods as the structural fix

EDUCATION

University of Waterloo Waterloo, ON
Candidate for B.ASc. in Honours Mechanical Engineering *Sep 2025 - Apr 2030 (expected)*

- **Relevant Courses**: Mechanics of Deformable Solids, Control of Material Properties, Electromechanical Devices
- **Scholarships**: President's Scholarship of Distinction